

BEE 4750 Homework 5: Mixed Integer and Stochastic Programming

Name: Moho Goswami

ID: mg2297

Due Date

Thursday, 12/04/24, 9:00pm

Overview

Instructions

- In Problem 1, you will use mixed integer programming to solve a waste load allocation problem.
- In Problem 2, you will formulate a stochastic optimization problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
Activating project at `c:\Users\mohog\hw5-mohog`
Installed Measures      v0.3.3
Installed HiGHS_jll     v1.12.0+0
Installed GR_jll        v0.73.18+0
```

```

Installed PlotUtils          v1.4.4
Installed OpenSSL            v1.6.0
Installed OpenBLAS32_jll     v0.3.29+0
Installed MutableArithmetics v1.6.7
Installed Pango_jll          v1.57.0+0
Installed FFMPEG              v0.4.5
Installed StaticArraysCore    v1.4.4
Installed JSON                v1.3.0
Installed DataStructures      v0.19.3
Installed GraphRecipes        v0.5.15
Installed METIS_jll           v5.1.3+0
Installed StatsBase           v0.34.8
Installed FFMPEG_jll          v8.0.0+0
Installed StableRNGs          v1.0.4
Installed HiGHS               v1.20.1
Installed ForwardDiff         v1.3.0
Installed StructUtils         v2.6.0
Installed JuMP                v1.29.3
Installed GR                  v0.73.18
Precompiling project...
11607.4 ms Measures
5498.5 ms StableRNGs
5546.1 ms StaticArraysCore
5640.9 ms StructUtils
8191.7 ms DataStructures
3204.3 ms METIS_jll
3179.7 ms OpenBLAS32_jll
6637.1 ms ChainRulesCore
4222.0 ms ColorVectorSpace → SpecialFunctionsExt
8963.4 ms OpenSSL
4816.6 ms DiffResults
10610.5 ms Pango_jll
2641.7 ms StructUtils → StructUtilsTablesExt
13225.8 ms FFMPEG_jll
3560.6 ms SortingAlgorithms
27031.7 ms MutableArithmetics
5604.7 ms HiGHS_jll
4079.4 ms ChainRulesCore → ChainRulesCoreSparseArraysExt
16492.0 ms JSON
6445.1 ms LogExpFunctions → LogExpFunctionsChainRulesCoreExt
7692.3 ms SpecialFunctions → SpecialFunctionsChainRulesCoreExt
6640.9 ms FFMPEG
32769.1 ms StaticArrays

```

16082.5 ms	ForwardDiff
44188.4 ms	PlotUtils
11670.4 ms	StatsBase
7304.7 ms	BenchmarkTools
22460.1 ms	GR_jll
9459.6 ms	ArnoldiMethod
11684.6 ms	StaticArrays → StaticArraysChainRulesCoreExt
3414.4 ms	Adapt → AdaptStaticArraysExt
3457.1 ms	StaticArrays → StaticArraysStatisticsExt
6525.5 ms	ForwardDiff → ForwardDiffStaticArraysExt
18415.9 ms	PlotThemes
20201.8 ms	RecipesPipeline
67588.2 ms	HTTP
7990.8 ms	Interpolations
7808.7 ms	GeometryTypes
47630.1 ms	GeometryBasics
6012.6 ms	Interpolations → InterpolationsForwardDiffExt
17550.2 ms	Graphs
17458.5 ms	GR
6605.0 ms	NetworkLayout
5830.7 ms	NetworkLayout → NetworkLayoutGraphsExt
8368.0 ms	GraphRecipes
137536.9 ms	DataFrames
7178.3 ms	Latexify → DataFramesExt
129874.0 ms	MathOptInterface
7744.7 ms	MathOptIIS
28885.2 ms	HiGHS
54283.1 ms	JuMP
179673.3 ms	Plots
21174.3 ms	Plots → GeometryBasicsExt

53 dependencies successfully precompiled in 438 seconds. 199 already precompiled.

```

using JuMP
using HiGHS
using DataFrames
using GraphRecipes
using Plots
using Measures
using MarkdownTables

```

Problems (Total: 30 Points)

Problem 1 (24 points)

Three cities are developing a coordinated municipal solid waste (MSW) disposal plan. Three disposal alternatives are being considered: a landfill (LF), a materials recycling facility (MRF), and a waste-to-energy facility (WTE). The capacities of these facilities and the fees for operation and disposal are provided below.

- **LF:** Capacity 200 Mg, fixed cost \$2000/day, tipping cost \$50/Mg;
- **MRF:** Capacity 350 Mg, fixed cost \$1500/day, tipping cost \$7/Mg, recycling cost \$40/Mg recycled;
- **WTE:** Capacity 210 Mg, fixed cost \$2500/day, tipping cost \$60/Mg;

Transportation costs are \$1.5/Mg-km, and the relative distances between the cities and facilities are provided in the table below.

City/Facility	Landfill (km)	MRF (km)	WTE (km)
1	5	30	15
2	15	25	10
3	13	45	20
LF	-	32	18
MRF	32	-	15
WTE	18	15	-

The fixed costs associated with the disposal options are incurred only if the particular disposal option is implemented. The three cities produce 100, 90, and 120 Mg/day of solid waste, respectively, with the composition provided in the table below.

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Food Wastes	15	8	0
Paper & Cardboard	40	7	55
Plastics	5	5	15
Textiles	3	10	10
Rubber, Leather	2	15	0
Wood	5	2	30
Yard Wastes	18	2	40
Glass	4	100	60
Ferrous	2	100	75
Aluminum	2	100	80

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Other Metal	1	100	50
Miscellaneous	3	70	0

The information in the above table will help you determine the overall recycling and ash fractions. Note that the recycling residuals, which may be sent to either landfill or the WTE, have different ash content than the ash content of the original MSW. You will need to determine these fractions to construct your mass balance constraints.

Reminder: Use `round(x; digits=n)` to report values to the appropriate precision!

Problem 1.1

Based on the information above, calculate the overall recycling and ash fractions for the waste produced by each city.

City 1

- 100 Mg/day solid waste
- Food waste: $15\% \cdot 100 = 15$ Mg/day
- Paper & Cardboard: $40\% \cdot 100 = 40$ Mg/day
- Plastics: $5\% \cdot 100 = 5$ Mg/day
- Textiles: $3\% \cdot 100 = 3$ Mg/day
- Rubber/leather: $2\% \cdot 100 = 2$ Mg/day
- Wood: $5\% \cdot 100 = 5$ Mg/day
- Yard wastes: $18\% \cdot 100 = 18$ Mg/day
- Glass: $4\% \cdot 100 = 4$ Mg/day
- Ferrous: $2\% \cdot 100 = 2$ Mg/day
- Aluminum: $2\% \cdot 100 = 2$ Mg/day
- Other metal: $1\% \cdot 100 = 1$ Mg/day
- Misc: $3\% \cdot 100 = 3$ Mg/day

Overall recycling fraction

$$0\% \cdot 15 + 55\% \cdot 40 + 15\% \cdot 5 + 10\% \cdot 3 + 0\% \cdot 2 + 30\% \cdot 5 + 40\% \cdot 18 + 60\% \cdot 4 + 75\% \cdot 2 + 80\% \cdot 2 + 50\% \cdot 1 + 0\% \cdot 3$$

$$= 37.50\%$$

Overall ash fraction

$$8\% \cdot 15 + 7\% \cdot 40 + 5\% \cdot 5 + 10\% \cdot 3 + 15\% \cdot 2 + 2\% \cdot 5 + 2\% \cdot 18 + 100\% \cdot 4 + 100\% \cdot 2 + 100\% \cdot 2 + 100\% \cdot 1 + 70\% \cdot 3$$
$$= 16.41\%$$

City 2

- 90 Mg/day solid waste
- Food waste: $15\% \cdot 90 = 13.5$ Mg/day
- Paper & Cardboard: $40\% \cdot 90 = 36$ Mg/day
- Plastics: $5\% \cdot 90 = 4.5$ Mg/day
- Textiles: $3\% \cdot 90 = 2.7$ Mg/day
- Rubber/leather: $2\% \cdot 90 = 1.8$ Mg/day
- Wood: $5\% \cdot 90 = 4.5$ Mg/day
- Yard wastes: $18\% \cdot 90 = 16.2$ Mg/day
- Glass: $4\% \cdot 90 = 3.6$ Mg/day
- Ferrous: $2\% \cdot 90 = 1.8$ Mg/day
- Aluminum: $2\% \cdot 90 = 1.8$ Mg/day
- Other metal: $1\% \cdot 90 = 0.9$ Mg/day
- Misc: $3\% \cdot 90 = 2.7$ Mg/day

Overall recycling fraction

$$0\% \cdot 13.5 + 55\% \cdot 36 + 15\% \cdot 4.5 + 10\% \cdot 2.7 + 0\% \cdot 1.8 + 30\% \cdot 4.5 + 40\% \cdot 16.2 + 60\% \cdot 3.6 + 75\% \cdot 1.8 + 80\% \cdot 1.8 + 50\% \cdot 0.9 + 0\% \cdot 2.7$$
$$= 33.975\%$$

Overall ash fraction

$$8\% \cdot 13.5 + 7\% \cdot 36 + 5\% \cdot 4.5 + 10\% \cdot 2.7 + 15\% \cdot 1.8 + 2\% \cdot 4.5 + 2\% \cdot 16.2 + 100\% \cdot 3.6 + 100\% \cdot 1.8 + 100\% \cdot 1.8 + 100\% \cdot 0.9 + 70\% \cdot 2.7$$
$$= 14.769\%$$

City 3

- 120 Mg/day solid waste
- Food waste: $15\% * 120 = 18 \text{ Mg/day}$
- Paper & Cardboard: $40\% * 120 = 48 \text{ Mg/day}$
- Plastics: $5\% * 120 = 6 \text{ Mg/day}$
- Textiles: $3\% * 120 = 3.6 \text{ Mg/day}$
- Rubber/leather: $2\% * 120 = 2.4 \text{ Mg/day}$
- Wood: $5\% * 120 = 6 \text{ Mg/day}$
- Yard wastes: $18\% * 120 = 21.6 \text{ Mg/day}$
- Glass: $4\% * 120 = 4.8 \text{ Mg/day}$
- Ferrous: $2\% * 120 = 2.4 \text{ Mg/day}$
- Aluminum: $2\% * 120 = 2.4 \text{ Mg/day}$
- Other metal: $1\% * 120 = 1.2 \text{ Mg/day}$
- Misc: $3\% * 120 = 3.6 \text{ Mg/day}$

Overall recycling fraction

$$0\% \cdot 18 + 55\% \cdot 48 + 15\% \cdot 6 + 10\% \cdot 3.6 + 0\% \cdot 2.4 + 30\% \cdot 6 + 40\% \cdot 21.6 + 60\% \cdot 4.8 + 75\% \cdot 2.4 + 80\% \cdot 2.4 + 50\% \cdot 1.2 + 0\% \cdot 3.6$$

$$= 45.3\%$$

Overall ash fraction

$$8\% \cdot 18 + 7\% \cdot 48 + 5\% \cdot 6 + 10\% \cdot 3.6 + 15\% \cdot 2.4 + 2\% \cdot 6 + 2\% \cdot 21.6 + 100\% \cdot 4.8 + 100\% \cdot 2.4 + 100\% \cdot 2.4 + 100\% \cdot 1.2 + 70\% \cdot 3.6$$

$$= 19.692\%$$

Problem 1.2

What are the decision variables for your optimization problem? Provide notation and variable meaning.

Problem 1.3

Formulate the objective function. Make sure to include any needed derivations or justifications for your equation(s).

Problem 1.4

Derive all relevant constraints. Make sure to include any needed justifications or derivations.

Problem 1.5

Find the optimal solution (using JuMP to solve the problem). Report the optimal objective value.

Problem 1.6

Draw a diagram showing the flows of waste between the cities and the facilities. Which facilities (if any) will not be used? Does this solution make sense?

Problem 2 (6 points)

Consider a two-period economic dispatch problem, based on the multi-period example from Lecture 14 (on 10/29). The generator data, including ramping constraints for each generator, is provided in ‘data/generators.csv.’ In period 1, the demand is $d_1 = 1100\text{MW}$. In period 2, the demand is projected to be $d_2 = 1200\text{MW}$, but there is a 25% probability that it is \$1500. In the first period, the solar capacity factor is 0.9 and the wind capacity factor is 0.45, but in the second period, there is some uncertainty: the forecasted solar and wind capacity factors are 0.95 and 0.4, respectively, but there is a 30% probability that they are 0.75 and 0.5. Your goal is to identify how to dispatch your generators to minimize the cost of meeting demand.

Problem 2.1

Draw a scenario tree for this problem.

Problem 2.2

Formulate a stochastic linear program for this problem based on your scenario tree from Problem 2.1 and the data in `data/generators.csv`.

Data from CSV file

Plant	Resource	$P_{\min}$	$P_{\max}$	VarCost	Ramp
Biomass	Biomass	0	100	5	100
Hydroelectric	Hydroelectric	0	500	0	500
Geothermal	Geothermal	0	0	400	400
NG CCGT	NG CCGT	220	500	23	100
NG CT	NG CT	100	250	38	200
Wind	Wind	0	300	0	300
Solar	Solar	0	500	0	500

References

List any external references consulted, including classmates.